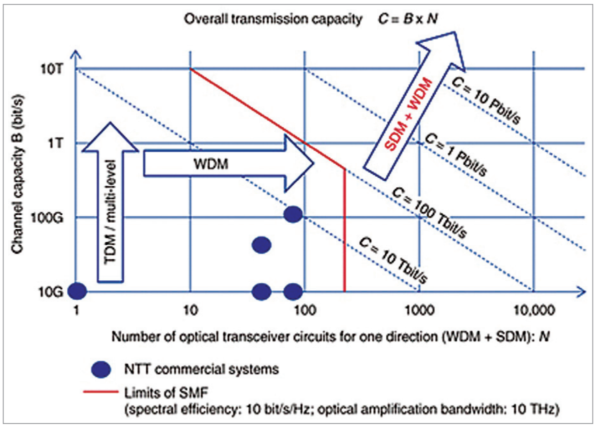

Fig. 6: Integration of SDM and WDM for high-capacity optical transport network (Source: https://www.ntt-review.jp)

Higher cost of manufacturing. MCFs require optimising multiple parameters, making them more expensive than SCFs. A 7-core MCF can cost double the cost of seven single-core fibres

Need for amplification techniques. MCFs and multiband networks require compatible optical amplifiers to fully utilise these technologies

Limited transmission distance. MCFs are primarily used for shorter distances, such as within buildings, due to higher signal attenuation over long distances compared to SCF

Future with the integration of multicore fibre and multiband optical networks

According to an article on SDM by NTT, Japan, existing SMF-based commercial systems have reached a channel capacity of up to 100Gbit/s using time division multiplexing (TDM), with approximately 100 multiplexed optical transceiver circuits, yielding a total transmission capacity of 100Tbit/s.

Current technology has reached its maximum capacity, necessitating new research and development. Integrating SDM and WDM presents a promising solution.

Combining MCF with multiband optical networks enables the use of SDM (from MCF) and WDM (from multiband networks), increasing system transmission capacity (Fig. 6) at the cost of requiring more optical transceiver circuits.

Data demand is escalating, and current technology is approaching its limits. To meet future bandwidth needs, integrating MCF and multiband optical networks is crucial. These technologies will redefine fibre-optic communications, boosting connectivity and increasing data transmission capacity while overcoming infrastructure challenges. EFY